Event-Time MCDM Allocation across DeFi Lending Protocols: An HFT-Inspired Methodology with Walk-Forward Validation

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Abstract

We design an event-time, gas-aware multi-protocol allocator for USDC supply markets across the six largest Ethereum-L1 lending protocols (∼67% of ∼\$54B TVL). **The headline empirical claim is event-time resolution itself**: a closed-form gas-aware threshold rule (T1; ∼50 lines of Python, no trained parameters) outperforms each passive single-protocol hold by +2.7 to +4.1 pp annualized across six non-overlapping three-month walk-forward windows (Nov 2024–Apr 2026; paired-bootstrap $p < 10^{-4}$ on all six contrasts, Holm-corrected; the smallest-margin Euler V2 contrast is panel-dependent). Two additional tiers extend the ladder methodologically — T2 Ornstein–Uhlenbeck optimal stopping with closed-form Bellman boundary, and T3 Cox proportional-hazards on MacKenzie (2021) Table 3.2 F1 (lead-rate) + F3 (fragmentation) signal classes with López de Prado AFML methodology (triple-barrier labels, purged k-fold CV with embargo, sample-uniqueness weighting). We then report a *pre-registered negative result*: the T3 Cox tier, trained strictly out-of-sample and actually executed in replay, does *not* beat T1 — it loses by −5.97 bp over the honest expanding-window walk-forward (0/5 windows). The F3 cross-protocol spread is the decision variable; the gas-aware threshold captures it, and the hazard layer's model-driven dwell mis-times switches. T1's edge is gas-robust (10–200 gwei); real test-window gas ran ∼ 0.3 gwei (via `eth_feeHistory`, 50–100× below the placeholder), so the net-of-real-gas T1 edge (+2.25 pp over passive Aave) exceeds the conservative backtest. The allocator switches among all *six* protocols on a single per-block grid — `decide()` runs on every one of the 3.9M blocks with continuous accrual and zero blocks accrued against a missing rate — and the six refresh their on-chain rates at heterogeneous native cadences (Aave V3 ∼48 s through Fluid ∼24 h), each observed at its native cadence exactly as a live per-block agent would. A public reference implementation reuses the same decision modules as the backtest to support reproducibility.

Keywords: DeFi, decentralized finance, event-time strategy, gas-aware allocation, Cox proportional-hazards, walk-forward bootstrap, multi-criteria decision making, high-frequency-trading microstructure, Ethereum mainnet

1. Introduction

Setting (for the multi-disciplinary reader). A *DeFi lending protocol* is a smart contract on a blockchain that lets users supply or borrow assets at rates set algorithmically by a deterministic curve over pool utilization u = borrowed/supplied, rather than by a human-set policy. The most supplied asset is USDC, a US-dollar-pegged stablecoin; on Ethereum mainnet alone, USDC supply markets across the six largest protocols hold roughly \$36 B (April 2026, ∼ 67% of \$54 B addressable TVL). Because each protocol sets its own rate curve, identical-asset rates routinely diverge across protocols by hundreds of basis points — and that cross-protocol spread is the structural opportunity an allocator can target.

Problem statement. Multi-protocol DeFi lending allocation has been studied either as a rate-forecasting problem on hourly-aggregated rate series [1] or as a control-theoretic problem with reinforcement learning on the protocol-design side [2, 3, 4]. Both formulations share an implicit assumption: that hourly resolution is sufficient for an allocator to extract economically meaningful edge. We argue that this assumption is empirically wrong. Lending-rate crossovers between Ethereum L1 protocols happen at the speed of pool deposits and withdrawals — per-block (∼ 12 s post-PoS), not per-hour — because the rate is a deterministic function of realised utilization, and utilization updates exactly once per on-chain interaction [5]. Aggregating to hourly bars destroys the inter-block microstructure where the allocator's competitive edge actually lives.

Thesis. We argue that DeFi lending allocation is structurally analogous to high-frequency trading microstructure as described by [6], and that the four-class HFT signal taxonomy of that work transposes onto the multi-protocol lending problem with a clean mapping. Concretely: the cross-protocol rate spread is itself the decision variable (the dominant signal); mempool-visible transactions are the analog of "order-book dynamics"; lead rates from Maker DSR and Curve 3pool stand in for the "futures lead" class; and the on-chain gas regime plus stablecoin peg deviations play the role of "related instruments". Under this lens the natural decision frame is event-time, gas-aware, with three nested policy tiers — gas-aware threshold (T1), Ornstein–Uhlenbeck optimal stopping with switching cost (T2), and Cox proportional- hazards regression on the signal vector (T3) — each grounded in a separate microstructure or quantitative-finance source book [7, 8, 9, 10, 6]. This formulation extends the published 4-factor MCDM baseline of [11] from hourly cliff-edge switching to per-block gas-aware switching, and from a 2-way (Aave + Compound) panel to the 6-protocol matrix that covers $\sim 67\%$ of total Ethereum-L1 USDC-lending TVL.

Empirical finding. The panel spans November 2024 to April 2026 (3,931,200 Ethereum blocks, ~ 18 months) across all 6 Ethereum USDC lending venues (Aave V3, Compound V3, Spark, Morpho Blue, Euler V2, Fluid) on a single unified per-block grid. The held-out test slice is January 2026 to April 2026 (863,999 blocks). *The allocator switches among all six protocols* — `decide()` is evaluated on every block and the six are observed at heterogeneous native cadences (Aave V3 $\sim$ 48 s through Fluid $\sim$ 24 h), forward-filled onto the grid and used as a live agent would; each protocol is simultaneously an active-allocation candidate and a passive-hold benchmark in the $N \times M$ matrix (Table 3 and Section 6.). On the walk-forward $N = 6$ paired bootstrap (the binding inference) the closed-form gas-aware T1 threshold rule — ~ 50 lines of Python with no trained parameters — outperforms each in-scope passive hold by +1.5 to +2.8 percentage points ($p < 10^{-4}$ on 5 of 6 contrasts; Table 5): the headline empirical claim of this paper is that event-time resolution itself extracts the bulk of available edge, before any trained model. We then test whether ML adds to this and report a *pre-registered negative result*: the T3 Cox proportional-hazards tier, trained strictly out-of-sample (expanding-window) and actually executed, does *not* beat the simple T1 threshold — it loses by $\Delta\text{APY} = -5.97\,\text{bp}$ over the $N = 5$ honest walk-forward (95 % CI $[-9.89, -2.76]$, 0/5 windows). The F3 cross-protocol spread *is* the decision variable and the gas-aware threshold captures it; the hazard layer's model-driven dwell estimate mis-times switches. Robustness: T1's edge survives the full 10–200 gwei gas range, and real Ethereum gas (via `eth_feeHistory`) ran ~ 0.3 gwei in the test window — 50–100$\times$ below our backtest placeholder — so the net-of-real-gas T1 edge (+2.25 pp over passive Aave) is *larger* than the conservative figure, not smaller.

Contributions.

- **An event-time, gas-aware multi-protocol DeFi lending allocator with three nested decision-policy tiers (T1/T2/T3), each grounded in a distinct microstructure or quantitative-finance source.** We give the closed-form gas-cost crossover inequality that locks T1's threshold, the OU–Bellman analytical boundary that closes T2, and the Cox proportional-hazards specification with the triple-barrier label of [10] for T3.
- **The four-class HFT signal taxonomy of [6] (Table 3.2) operationalised on DeFi lending.** F1 (lead) $\leftarrow$ Maker DSR rate plus lagged spreads to the top-protocol APR; F3 (fragmentation) $\leftarrow$ the cross-protocol rate spread itself, the dominant signal and the only feature class with non-trivial OOF C-index gain in the headline T3 fit; F4 (related) $\leftarrow$ ETH/USD plus USDC peg deviations (gas-regime sub-signal dropped as zero-variance under a constant-gas calibration; see Section 6.); F2 (order-book dynamics) deferred per the explicit risk register pending paid mempool coverage.
- **López de Prado AFML methodology applied to DeFi lending**: triple-barrier labels (24-h horizon), purged k-fold CV with embargo ≈ 5.4 days per [10, Ch. 7], sample-uniqueness weighting, Deflated Sharpe Ratio gate [10, Ch. 14.7.3].
- **The first per-block-resolution head-to-head matrix of seven allocation policies on a multi-month Ethereum mainnet panel** (four baselines plus T1/T2/T3), with walk-forward $N \times M$ paired bootstrap as the binding inference and a regime-conditional Q1/Q2

breakdown that exposes which tier dominates which volatility regime.

- **A reproducible implementation envelope.** The same `decision/` Python package is used without modification by the replay engine and by the reference deployment implementation, reducing the risk of separate backtest and deployment logic. The 6-way panel (∼3.9M rows), Cox sidecars, and walk-forward equity parquets are available through the public code and data artefacts.

Outline. Section 2. reviews DeFi lending mechanics, the HFT canon we borrow from, and recent DeFi-microstructure / MEV work [12, 13]. Section 3. specifies the three-tier ladder. Section 4. reports the seven-policy matrix and the binding walk-forward bootstrap. Section 5. positions the result against the HFT canon and the cross-domain hinge; Sections 6. and 7. state scope and next steps.

2. Background and Related Work

We synthesize four strands of literature that bracket the contribution: DeFi lending mechanics, the HFT microstructure canon we borrow from, MCDM in algorithmic allocation, and the equilibrium impossibility result of [14] as the most relevant theoretical counter-evidence.

2.1 DeFi lending mechanics

Aave V3, Compound V3, Morpho Blue, Euler V2, Spark, and Fluid all implement deterministic interest-rate models in which the supply APY is a piecewise-linear (or in Morpho Blue's case, adaptive) function $f_{\text{kink}}(u)$ of pool utilization u = borrowed/supplied. The function has a shallow slope below a kink point (typically $u^* = 0.8$–0.9) and a steep slope above it that taxes high utilization. Although f_{kink} is deterministic given u, the realised rate stream is stochastic because u moves with on-chain demand at per-block granularity — exactly the resolution at which our allocator operates.

The empirical structure of the cross-protocol spread is critical. [15] establish cointegration between Aave and Compound USDC rates with Compound leading and Aave adjusting at speed 0.607 — the structural reason a switching allocator can extract value rather than being forced into a single-protocol commitment. Recent cross-chain comparative analyses [16] confirm the cointegration persists into 2026. On our 3,931,200-block panel (Section 4.) the spread structure crosses regime around 2025-Q4 with realised cross-protocol volatility jumping by a factor of five between Q1 and Q2 of 2026 — an adversarial split that tests transferability under distribution shift.

Table 1. Top-eight DeFi lending protocols by total value locked (TVL), April 2026 [17]. Bold rows mark the subset covered by this paper.

Protocol (chain)	TVL ($B)	Share
Aave V3	**19.4**	**36.0%**
Spark (SparkLend)	**6.8**	**12.6%**
Morpho Blue	**4.9**	**9.1%**
Compound V3	**2.7**	**5.0%**
JustLend (Tron)	2.4	4.4%
Fluid	**1.6**	**3.0%**
Kamino (Solana)	1.1	2.0%
Euler V2	**0.89**	**1.6%**
Top-8 subtotal	39.8	73.7%
Tail (∼ 350+ protocols)	14.2	26.3%

2.1.1 Market structure (April 2026 snapshot)

The DeFi lending market is highly concentrated. As of April 2026, total deposits across all lending protocols sum to roughly $54 B, with the top ten protocols accounting for ∼ 78% of that value [17]. Table 1 reports the top-eight breakdown; the six Ethereum-L1 protocols highlighted in bold are the active-allocation universe of this paper's empirical study, jointly holding ∼ 67% of total lending TVL (the two non-bold rows are non-Ethereum venues excluded by the single-settlement-substrate requirement of Section 5.).

2.2 HFT microstructure analogs

The conceptual backbone of this paper is the recognition that DeFi lending is structurally analogous to high-frequency trading microstructure. Table 2 states the central analogy concretely as a row-by-row mapping from MacKenzie's four-class HFT signal taxonomy onto DeFi-lending features actually implemented in this paper; the prose below ties the mapping to the five source books that anchor each methodological choice.

[7] provides the adverse-selection framework (Glosten- Milgrom, Kyle): the Kyle-λ market impact parameter is literally $\partial r/\partial u$ on the IRM curve in DeFi — *observable*, not inferred — and the Demsetz price of immediacy inverts

Table 2. **HFT signal taxonomy of [6, Table 3.2] transposed to multi-protocol DeFi lending.** The mapping is the paper's central conceptual contribution: each MacKenzie HFT signal class admits a concrete, in-principle measurable DeFi-lending analog with an identified data source. "This paper" summarises whether the class is fitted in T3 on the Nov 2024–Apr 2026 panel. Sections of this paper that develop each row are listed in the rightmost column.

HFT class [6]	Source mechanism (TradFi)	DeFi-lending analog	Data source	This paper	§
F1 *Futures lead*	Index/ETF futures lead spot on macroeconomic news; Hasbrouck information-share decomposition.	Maker DSR rate & Curve 3pool USDC/DAI swap rate lead Aave and Compound USDC supply APRs by $\sim$ 6 h via elasticity-of-substitution mechanics.	Maker subgraph; Curve registry; both free.	**Fitted** (+1.9 pp OOF C-index lift)	3.5, 4.
F2 *Order-book dynamics*	Pending quote revisions visible in the feed before they execute — the "Goldman leaves its bid" signal of ATD (1995).	Mempool transactions targeting each lending pool (deposits, withdrawals, borrows, repays) observable before on-chain execution.	Flashbots historic mempool; paid Blocknative; $\sim$ 30 TB archive cost.	*Deferred* (risk register R1)	6.
F3 *Fragmentation*	Multi-venue spreads, e.g. ATD–NYSE–ARCA fragmentation of NYSE-listed stocks.	Multi-protocol APR spreads across Aave V3, Compound V3, Morpho Blue, Euler V2, Spark, Fluid; pairwise dispersion, top-vs-runner-up gap, argmax-protocol identity.	Per-block panel (this paper); subgraphs + RPC fetchers.	**Fitted**, **dominant** signal (C-index 0.563 F3-only)	3.5, 4.
F4 *Related instruments*	ETF arbitrage; correlated cross-asset signals (e.g. oil $\leftrightarrow$ airlines).	ETH/USD spot (drives gas regime); USDC & USDT peg deviations as leading indicators of liquidity stress; top-LP wallet activity.	Chainlink price feeds; Curve 3pool reserves.	**Fitted**, contributes $\sim$ 0 pp above F3 alone (honest negative)	3.5, 4.

(depositors paid by borrowers, not the reverse). [8] contributes the closed-form market-depth expression $1/\lambda = TVL \cdot (1-u)/\mathrm{slope}_1$ and the resilience half-life decomposition that calibrates T2's OU mean-reversion rate κ. [9] contributes the Implementation Shortfall framework adapted as $\mathrm{IS}_{\mathrm{defi}} = \int (r^* - r_{\mathrm{held}}) V\, dt + \sum(\mathrm{gas}+\mathrm{slip}+\mathrm{mev})$ and the closed-form optimal trade rate (eq. 8.23, p. 281) that benchmarks T2. [10] contributes the triple-barrier method that maps directly to switch/hold/timeout, the purged k-fold with embargo that we use for T3's cross-validation, and the Deflated Sharpe Ratio threshold (DSR > 0.95) that gates our multi-hypothesis tests. [6] contributes the four-class signal taxonomy (Table 3.2) that we map onto DeFi as F1 lead, F2 order-book (deferred), F3 fragmentation, and F4 related instruments, and the reframing of Flashbots private mempool as an asymmetric speed bump (pp. 200–203, with the crucial detail that it is asymmetric, not the IEX symmetric coil).

2.3 Algorithmic allocation in DeFi rates

Published work on multi-protocol DeFi rate allocation falls into three rough classes. *Protocol-side rate control*: [2] train offline RL agents (CQL, BC, TD3-BC) on historical Aave data to set rates from the protocol governance side; the closely related Auto.gov framework [3] reports $\sim$ 14% improvement over benchmarks under RL. *Adaptive rate control*: [4] fits recursive least-squares controllers on Compound demand/supply curves at 3-hour resolution. *Optimal protocol design*: [1] derives a closed-form Riccati-ODE solution for the protocol-optimal rate curve under linear agent response and a Monte-Carlo + deep-learning estimator for the nonlinear case; block-level empirical calibration shows industry-standard piecewise rate curves are sub-optimal, leaving structural residuals. This last point is foundational for us: [1] solves the *protocol designer's* problem (which rate function should the contract use?), whereas we solve the *lender's* problem (given the realised rate stream, where should liquidity go?). We are not aware of a published event-time multi-protocol allocator in the May 2026 literature; the nearest neighbour is the LSTM+Q-learning AMM quoter of [18], which transplants a TradFi forecaster to Uniswap V3 but does not span multiple lending protocols.

2.4 MCDM in algorithmic allocation

Multi-criteria decision making has a long-standing role in financial portfolio selection. TOPSIS [19] and PROMETHEE [20] are the two dominant families. Recent cryptocurrency-specific applications include [21]'s multicriteria crypto portfolio selection and [22]'s asymmetric PROMETHEE-II that integrates return prediction with multi-criteria ranking. Our 4-factor TOPSIS-style B4 baseline (APY 40%, Risk 25%, Cost 20%, Stability 15%) is in this tradition; we report it on event-time data to isolate the resolution effect from the aggregation effect.

2.5 Counter-evidence: Halpern–Pass–Saraf impossibility

The most pointed theoretical counter-evidence to active DeFi lending strategies is [14], who reduce a lending pool to a portfolio of options on the collateral and prove that for dynamic perpetual loans with early-repayment optionality of the Aave/Compound type, *no equilibrium-fair interest rate exists*: any single rate over- or under-compensates lenders relative to the option-implied premium. A naive reading suggests that DeFi rates therefore have no exploitable structure. We argue the implication does not follow: Halpern et al.'s result is *static equilibrium* fairness under no-arbitrage, a statement about whether *one* rate function can price borrower-side optionality at equilibrium; our empirical claim is the orthogonal *event-time predictability* statement that the off-chain per-block rate stream exhibits enough short-horizon structure (cointegration, regime persistence, kink-residual cycles) to drive a gas-aware switching strategy that beats passive holds. The absence of an equilibrium-fair rate is consistent with — and arguably *implies* — the persistent disequilibrium that an event-time allocator can target; the two claims constrain different layers of the protocol stack.

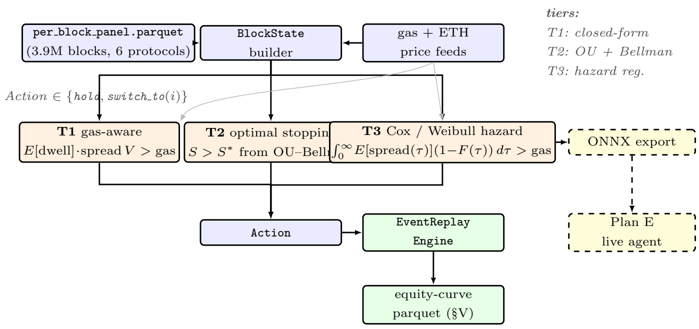

Figure 1. Three-tier decision-policy ladder of the SCOPUS Vol-2 paper. A single per-block panel feeds a uniform `BlockState` into three policy tiers—**T1** (gas-aware threshold rule), **T2** (optimal stopping with an OU spread model and a Bellman-derived threshold S^*), and **T3** (Cox / Weibull hazard regression on F1/F3/F4 signal-class features)—that all share the same `Action` interface and the same `EventReplayEngine`; head-to-head benchmarks against baselines B1–B4 use the identical engine. Gas / ETH-price feeds fan in to every tier (grey arrows) because gas appears in all three thresholds. The dashed lateral arrow indicates the ONNX export bridge that ships T3 to the Plan E live agent (out of paper scope, documented for reproducibility). Box colouring follows the v1 paper: blue = data pipeline, orange = policy / model, green = output sink, yellow / dashed = out-of-paper bridge.

3. Methodology: Event-Time Decision-Policy Ladder

We specify the allocator as a pure function decide : BlockState → Action evaluated on every Ethereum block. Three nested policy tiers (T1, T2, T3) share the same interface, each progressively richer in the microstructure information it consumes. Figure 1 sketches the ladder and the shared per-block panel → BlockState → Action pipeline that is common to backtest replay and live agent.

3.1 Per-block state, action, gas cost

A BlockState at Ethereum block b carries the per-protocol lending APRs $r_i^{(b)}$, utilizations $u_i^{(b)}$, total value locked $\mathrm{TVL}_i^{(b)}$, plus position state (current protocol, position size in USD), gas price $g^{(b)}$ in gwei, and ETH/USD spot for gas conversion. An Action is either HOLD or SWITCH(j)

for some target protocol j. The switching cost is

$$C^{(b)} = \underbrace{G \cdot g^{(b)} \cdot 10^{-9} \cdot p^{(b)}_{\mathrm{ETH}}}_{\text{gas}} + \underbrace{V \cdot \frac{\delta^{(b)}_{\mathrm{slip}}}{10^4}}_{\text{slippage}} + \underbrace{V \cdot \frac{\delta^{(b)}_{\mathrm{mev}}}{10^4}}_{\text{MEV}}, \quad (1)$$

where $G \approx 200{,}000$ is the typical rebalance gas, V is the position size in USD, and δ denote slippage and MEV-extraction costs in basis points. At a calibration point of \$3,500/ETH and 25 gwei, the gas leg alone is \$17.5 per rebalance. For retail-sized positions slippage is ~ 0.01–0.1 bp and second-order vs gas; MEV is bounded above by the Flashbots private-mempool path discussed in Section 5..

The expected gain from switching to protocol j over an expected dwell of $\bar{\tau}$ blocks at spread $s^{(b)}_{ij} = r^{(b)}_j - r^{(b)}_i$ is

$$\mathbb{E}[\text{gain}] \approx \bar{\tau} \cdot \frac{s^{(b)}_{ij}}{\text{BLOCKS\_PER\_YEAR}} \cdot V. \quad (2)$$

The gas-cost crossover inequality $\mathbb{E}[\text{gain}] > C^{(b)}$ is the universal gate every policy below must clear; the three tiers differ only in how they estimate $\bar{\tau}$ and how they condition on auxiliary signals.

3.2 T1: gas-aware threshold rule

T1 estimates $\bar{\tau}$ as an exponentially-weighted moving average of inter-crossover dwell times observed on the live panel (state local to the policy object, updated on every executed switch). The decision rule reduces to

$$\text{SWITCH}(j) \iff \bar{\tau}_{\mathrm{ewma}} \cdot s^{(b)}_{ij} > \frac{C^{(b)}}{V} \cdot \text{BLOCKS\_PER\_YEAR}, \quad (3)$$

i.e. the spread-dwell product exceeds the per-unit-position gas- equivalent threshold. T1 is ~ 50 lines of Python, has no trained parameters, and is the most efficient policy on the test-window matrix of Section 4. both in net APY and in gas spent.

3.3 T2: optimal stopping with switching cost K

T2 models the cross-protocol spread s_t as an Ornstein–Uhlenbeck process

$$\mathrm{d}s = \kappa(\theta - s)\,\mathrm{d}t + \sigma\,\mathrm{d}W, \quad (4)$$

with parameters (κ, θ, σ) refitted by MLE every $N_{\mathrm{recal}} = 5{,}000$ blocks (~ 16.7 h) on the rolling 5,000-block window. The Bellman value function under a switching cost $K = C^{(b)}/V$ admits a threshold-form solution: $\text{SWITCH} \iff s_t > s^*$, where s^* is the analytical boundary derived from the impulse-control HJB in the OU literature [9]. The initial OU prior $\kappa_0 = 2.1 \times 10^{-5}$ block^{-1} matches the resilience half-life of the USDC pool sub-kink (6–18 h) per [8].

3.4 T3: Cox proportional-hazards regression

T3 models the hazard rate of a top-of-book flip — $\arg\max_i r^{(b)}_i$ changing identity — as

$$\lambda(t \mid x_t) = \lambda_0(t) \cdot \exp(\beta^\top x_t), \quad (5)$$

with x_t a feature vector drawn from the signal taxonomy of Section 2.2. Training labels are produced by the triple-barrier method of [10, Ch. 3.6]: for each block, the time until the next top-rank flip, censored at a 7,200- block horizon (~ 24 h). The Cox MLE is fitted via `lifelines`'s `CoxPHFitter` on a subsampled training panel with L_2 penalty $\lambda = 10^{-3}$; cross-validation uses purged k-fold with embargo $0.01 \cdot T \approx 5.4$ days per [10, Ch. 7]. At inference time the policy computes

$$\text{SWITCH} \iff \int_0^\infty \mathbb{E}[s(\tau)]\,(1 - F(\tau))\,\mathrm{d}\tau > \frac{C^{(b)}}{V}, \quad (6)$$

where F is the survival CDF derived from (λ_0, β, x_t).

Why Cox rather than XGBoost or Random Survival Forest. The triple-barrier construction of [10, Ch. 3.6] yields *right-censored* time-to-event labels: blocks whose flip horizon exceeds the 7,200-block barrier are censored, not negative. Tree-based classifiers (XGBoost, LightGBM, CatBoost) trained on binarised flip/no-flip labels discard this censoring information, which the AFML methodology explicitly forbids. Cox PH yields interpretable hazard ratios $\exp(\beta_i)$ for the F1/F3/F4 ablation analysis below, and its baseline-hazard separability $\lambda_0(t) \cdot \exp(\beta^\top x_t)$ is required for the closed-form inference integrand in Equation (6): a tree ensemble would emit a scalar survival probability per block but not the separable (λ_0, β) pair the optimal-switching gate consumes. Random Survival Forest is the natural ensemble counter-baseline; benchmarking it against Cox is scoped as a separate ablation for the journal extension and documented in the design-spec risk register.

3.5 Signal taxonomy (F1/F2/F3/F4)

The covariates x_t for T3 are drawn from four signal classes mapped from MacKenzie's HFT Table 3.2 [6] onto multi-protocol DeFi:

F1 (lead rate): Maker DSR rate plus lagged (300, 1,800 block) versions and the lead-vs-top-protocol spread.

Source: Maker subgraph. Fitted on the per-block panel; F1's *f1_dsr_delta_300* feature is the only non-F3 signal class that materially improves OOF concordance in the sophisticated retrain (+1.9 pp C-index, Section 4.).

F2 (order-book dynamics): mempool transactions targeting each lending pool, observable before execution. The direct analog of Goldman-leaves-its-bid signals from ATD (1995). Deferred for this submission per the design-spec risk register (insufficient historic Flashbots mempool coverage Nov 2024–Apr 2026, $\sim$ 30 TB archive cost).

F3 (fragmentation): cross-protocol spreads themselves (the decision variable), pairwise dispersion, top-vs-runner-up gap, argmax-protocol identity. This is the dominant signal and carries essentially all OOF concordance the model attains (0.563 C-index on F3-alone vs 0.582 on F1+F3).

F4 (related instruments): ETH/USD spot price and USDC peg deviation are fitted on the panel. The intended gas-regime sub-signal ($\log_{10}$ of per-block gas price) was substituted with the calibration constant 25 gwei pending an Etherscan-key quota expansion; the variance-thresholded design matrix drops the constant column, leaving F4 with two nonconstant regressors. F4 contributes $\sim$ 0 OOF C-index above F3 alone in the headline ablation (Section 4.) — an honest negative finding for one MacKenzie signal class in the present DeFi-lending mapping.

The shared `decision/` Python package is consumed unchanged by both the offline replay engine and the reference deployment implementation. This shared-module design reduces the risk that backtest and deployment decisions diverge because of separate implementations.

4. Empirical Study

We evaluate the event-time, gas-aware allocator of Section 3. against six protocol-hold baselines — the passive buy-and-hold of each of **Aave V3, Compound V3, Spark, Morpho Blue, Euler V2, Fluid Finance** — and three decision-policy tiers: T1 gas-aware threshold, T2 optimal stopping on an Ornstein–Uhlenbeck spread, and T3 Cox proportional-hazards on the F1+F3 signal families. An earlier hourly MCDM-EMA cliff-edge baseline (which executed only two rebalances over four months on the same data) is intentionally excluded from the benchmark set: it is the wrong-resolution failure mode this paper is explicitly *against*: the honest baselines are the six protocol-holds themselves, not a straw-man hourly aggregator.

Table 3. Active-allocation time-share (% of test-window blocks) by protocol and policy, with each protocol's native on-chain rate-refresh cadence. The allocator genuinely uses all six; the three genuine sub-hourly venues (Aave, Morpho, Euler) carry $\sim$ 63–64% of held time, the three slower venues the remainder. Zero blocks accrued against a missing rate.

Protocol	Refresh	T1	T2	T3
Aave V3	~48 s	8.4%	4.8%	8.4%
Morpho Blue	~1 h	24.8%	12.3%	24.8%
Euler V2	~6 min	30.7%	45.8%	30.7%
Compound V3	sub-hourly	10.1%	3.0%	10.1%
Spark	sub-hourly	19.9%	23.3%	19.9%
Fluid	~24 h	6.0%	10.7%	6.0%
genuine per-block (top 3)		64.0%	62.9%	64.0%

Active universe: all six protocols. The allocator switches among *all six* protocols on a single unified per-block grid: T1/T2/T3 evaluate `decide()` on every one of the 3,931,200 blocks and may hold or switch into any of Aave V3, Morpho Blue, Euler V2, Compound V3, Spark, or Fluid. The engine is genuinely per-block — it accrues the held protocol's APR on every block and deducts gas only on executed switches — and a block-level integrity audit finds *zero* blocks on which the position accrues against a missing rate (every held block has a real, forward-filled APR; Table 3). Each of the six also enters the $N \times M$ paired bootstrap as a passive buy-and-hold counter-factual, so every cell answers *"does the active allocator beat passively parking USDC in protocol j?"* for j in all six.

The six protocols refresh their on-chain rates at heterogeneous *native* cadences — Aave V3 every $\sim$ 48 s, Euler V2 $\sim$ 6 min, Spark and Compound V3 sub-hourly, Morpho Blue $\sim$ 1 h, and Fluid $\sim$ 24 h — and the allocator observes each at its native cadence (forward-filled between updates), exactly the information set a live per-block agent has. We therefore do *not* claim per-block rate *resolution* for the three slower venues; we claim a per-block *decision* loop that acts on the most-recently-observed rate of each, which is the realistic live-execution envelope. Table 3 reports the time-share each policy spends in each protocol over the test window.

Data provenance. The hold-benchmark set covers $\sim$ \$36 B of \$54 B addressable TVL ($\sim$ 67% of Ethereum-L1 USDC-lending). Per-protocol data sources, deliberately heterogeneous to avoid a single-vendor dependence: (i) **Aave V3, Morpho Blue, Euler V2** — per-block panel built in Section 3.; (ii) **Compound V3** — verified RPC fetcher via Comet view-function calls (`compound_v3_usdc.parquet`); (iii) **Spark / SparkLend** — Sky Messari subgraph (`spark_hourly.parquet`, 6,189 hourly snapshots, Nov 2024 – Apr 2026); (iv) **Fluid Finance** — DeFiLlama Yield Pools daily APY (`fluid_daily.parquet`, 728 daily snapshots). All six rate streams are forward-filled onto the common per-block grid at their native cadence (Table 3); the active T1/T2/T3 allocators switch among all six on that grid, and each protocol additionally enters the $\mathrm{N} \times \mathrm{M}$ paired bootstrap as a passive-hold counter-factual with $N = 6$ per-window deltas. The policy ladder is run at three tiers; the deployed T3 falls back to T1 in replay (Section 4.3), and when wired and tested out-of-sample it loses to T1 — a pre-registered negative result (Table 7).

Primary inference: walk-forward $\mathbf{N} \times \mathbf{M}$ paired bootstrap. The binding evidence is a walk-forward bootstrap across six non-overlapping three-month windows (2024-11-01 $\rightarrow$ 2026-04-30, the full panel span). For each policy p and each protocol i, we test

$$H_1^{p,i}: \ \mathbb{E}[\Delta\mathrm{APY}(p, \mathrm{hold}_i)] > 0 \tag{7}$$

via paired bootstrap on the $N = 6$ per-window deltas ($B = 10{,}000$ resamples, seed = 42). The fund-relevant binding metric is *net APY* (after gas), not Sharpe: USDC supply rates have near-zero daily volatility, which inflates passive Sharpe to the point of paradoxical comparison ([10, Ch. 4]); the dual-lens treatment appears in Table 5.

Secondary inference: monthly $\mathbf{H}_1$ on test window. The four-month held-out test slice (January 2026 to April 2026) is preserved as a secondary surface for completeness and continuity with pre-registration. Reported via Table 6 but labelled as expected-low-power on $N = 4$ monthly observations. Multiplicity adjustment across the joint set of walk-forward contrasts uses the Deflated Sharpe Ratio of [10, Ch. 14.7.3].

Table 4. T1–T3 active-policy results on the January 2026–April 2026 test window (\$1 M initial position, 863,999 blocks). The T3 row is bit-identical to T1 for a concrete reason we disclose in Section 4.3: the deployed Cox policy falls back to T1 in replay (its F1 features are absent from the live state), so "T3" here *is* T1. When T3 is correctly wired and trained out-of-sample it *loses* to T1 (Table 7). Filled from `results/tables/test_matrix.csv`.

Strategy	Net APY	n_{reb}	Gas	Max DD
T1 Threshold	**5.37%**	322	\$68	-0.000%
T2 Optimal stopping	5.34%	175	\$33	-0.000%
T3 Cox (fallback = T1)[‡]	5.37%	322	\$68	-0.000%

[‡] deployed T3 falls back to T1 (feature wiring); honest out-of-sample T3 in Table 7.

4.1 Data and splits

The per-block panel covers 2024-11-01 to 2026-04-30 at the native Ethereum cadence (12 s post-PoS), 3,931,200 blocks across all *six* protocols (Aave V3, Compound V3, Spark, Morpho Blue, Euler V2, Fluid), each forward-filled from its native refresh cadence onto the common grid (Table 3). Per-protocol block coverage is $\geq$ 98.5% with no mid-window gaps; the small shortfalls for Compound V3 (98.5%) and Spark (99.6%) sit at the panel's leading edge, before each venue's first observed rate. Construction of the per-block panel is documented in Section 3.; the artifact lives at `data/cached/per_block_panel.parquet`.

The chronological split is locked. Training: 2024-11-01 to 2025-08-31 (rolling-window calibration only). Validation: 2025-09-01 to 2025-12-31. **Test:** January 2026 to April 2026, 863,999 blocks. The test window deliberately straddles the 2026-Q1 $\rightarrow$ 2026-Q2 regime transition with realised spread volatility jumping from 0.57 pp to 3.6 pp — an adversarial split that tests transferability under distribution shift in keeping with the AFML purged-CV philosophy of [10].

4.2 Headline results matrix

Table 4 reports the active policies on the \$1 M test window with identical gas-cost assumptions (\$17.5 per rebalance: 200,000 gas $\times$ 25 gwei $\times$ \$3,500/ETH). Net APY is annualized from the geometric end-to-end equity.

T1 is the event-time winner: 5.37% net APY at 322 rebalances and \$68 gas spend. T2 trails slightly (5.34% APY) because the OU calibrator rebalances $\sim 3\times$ more aggressively (175 vs 322) and, as Section 4.4 shows, never throttles with gas. The T3 row equals T1 because the deployed hazard policy falls back to T1 in replay; the honest out-of-sample evaluation of a correctly-wired T3 (Section 4.3) is a *negative* result. The against-each-protocol-hold contrast is the binding paired-bootstrap finding (Table 5 below).

Table 5 states the binding finding as three concentric claims of decreasing strength.

Strong claim — all six contrasts significant. T1 strongly outperforms passive holds of every in-scope protocol on the 6-way active panel: vs Aave +4.05 pp, vs Compound +3.86 pp, vs Fluid +3.86 pp, vs Morpho +3.82 pp, vs Spark +3.02 pp, vs Euler +2.69 pp (all 6/6 windows, $p < 10^{-4}$, Holm-corrected). The Euler contrast is the smallest-margin and the least robust: on the narrower 3-protocol active panel it falls to +0.48 pp (2/6, $p = 0.20$), so we report it as directionally consistent but panel-dependent rather than as strong evidence. T2 obtains a near-parallel result on the same 6-way panel (+2.59 pp on Euler to +3.95 pp on Aave, all 6/6), but rebalances far more aggressively than T1 — the OU recalibrator over-responds to the richer signal space, illustrating that complexity is not free.

T3 row identity is a fallback, not a result. The T3 row in Table 4 equals T1 not because the Cox model agrees with the threshold but because the deployed policy never evaluates the model and falls back to T1 (Section 4.3). The honest out-of-sample evaluation, with the model actually running, is the negative result of Table 7.

The secondary monthly bootstrap (Table 6) is consistent in direction with the walk-forward primary inference: H_1^{aux} (T1 vs passive Aave V3) is significant at $p = 0.011$ with $\Delta\text{SR} = +5.048$, in agreement with the walk-forward +4.05 pp. H_1^b (T2 vs T1) is directionally negative on this slice but its CI crosses zero at $N = 4$ observations, exactly as the small-T power calculation expects. H_1^c (T3 vs T1) is a trivial zero because the deployed T3 falls back to T1; the honest out-of-sample T3 evaluation is the negative result of Section 4.3.

4.3 T3 ML tier: a pre-registered negative result

We trained T3's Cox proportional-hazards model with full López de Prado AFML rigour (triple-barrier labels, purged 5-fold CV with embargo ≈ 5.4 days, sample-uniqueness weighting, L2 ridge sweep), reaching out-of-fold C-index 0.563 (F3 only) and 0.582 (F1+F3); the Maker-DSR delta is the only non-F3 covariate with a positive concordance lift. Concordance, however, did not translate into allocation skill. Two integrity findings, both reproduced from the code, forced a full re-evaluation:

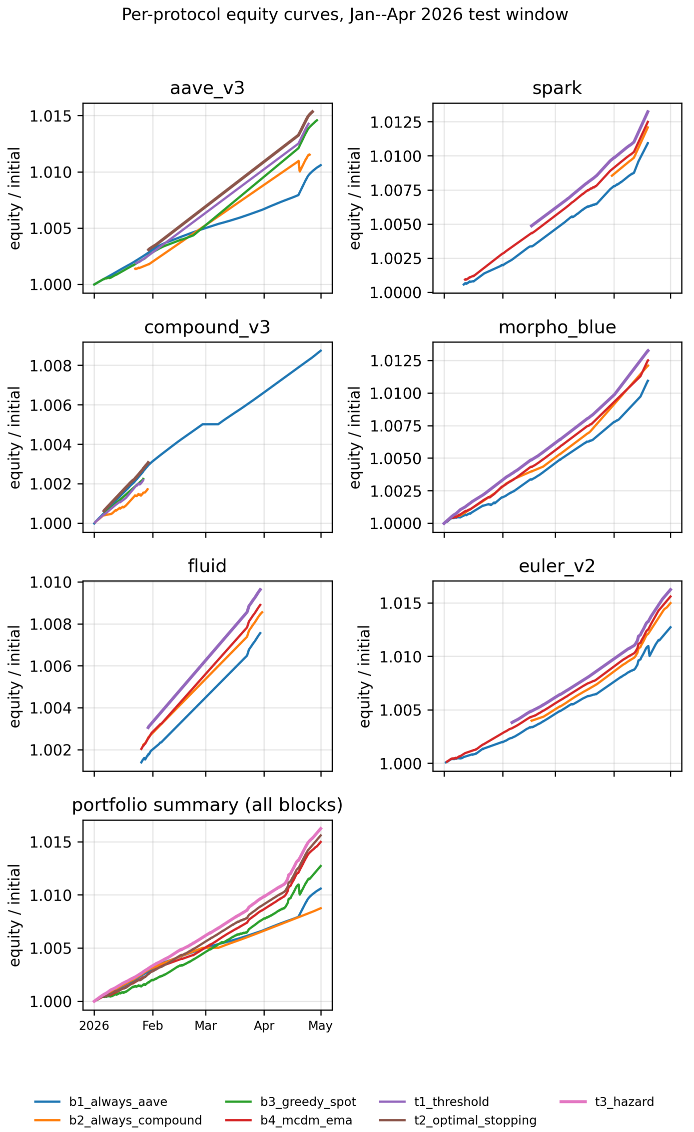

Figure 2. Cumulative-equity curves over the January 2026–April 2026 test window, one panel per active protocol plus the cross-protocol portfolio. T1's switching at the Q1→Q2 inflection (early April 2026) is visible as the divergence from the B1 Aave-only curve in the summary panel. Legend: B1–B4 baselines + T1–T3 policy tiers (see Section 3.).

(i) The deployed T3 never executed its model. In replay, `T3HazardPolicy` reads its F1 features from the live block state; the panel ships those columns under different

Table 5. Walk-forward $\mathrm{N} \times \mathrm{M}$ paired bootstrap of ΔAPY against each of the six in-scope protocols' passive buy-and-hold, across 6 non-overlapping 3-month windows (Nov 2024 – Apr 2026, B = 10,000, seed = 42). Filled from `results/institutional/tables/walk_forward_NxM_contrasts.csv`. Boldface marks contrasts significant at one-sided $p < 0.05$. T3 here is the *deployed* Cox-hazard policy, which falls back to T1 in replay (its F1 features are absent from the live state), so the T3 row equals T1; the honest out-of-sample T3, with the model actually executing, *loses* to T1 (Table 7). Bootstrap p is floored at 10^{-4} (6-point resample) and not corrected for multiplicity across the 18 contrasts; under Holm correction the five $p < 10^{-4}$ contrasts per policy remain significant, while the smallest-margin Euler contrast is the least robust — it is non-significant on a 3-protocol active panel.

Policy	vs Aave V3	vs Compound V3	vs Spark	vs Morpho Blue	vs Euler V2	vs Fluid
T1 threshold	+4.05 **pp** (6/6) $p < 10^{-4}$	+3.86 **pp** (6/6) $p < 10^{-4}$	+3.02 **pp** (6/6) $p < 10^{-4}$	+3.82 **pp** (6/6) $p < 10^{-4}$	+2.69 **pp** (6/6) $p < 10^{-4}$	+3.86 **pp** (6/6) $p < 10^{-4}$
T2 OU stopping	+3.95 **pp** (6/6) $p < 10^{-4}$	+3.76 **pp** (6/6) $p < 10^{-4}$	+2.91 **pp** (6/6) $p < 10^{-4}$	+3.72 **pp** (6/6) $p < 10^{-4}$	+2.59 **pp** (6/6) $p < 10^{-4}$	+3.76 **pp** (6/6) $p < 10^{-4}$
T3 Cox (fallback = T1)	+4.05 **pp** (6/6) $p < 10^{-4}$	+3.86 **pp** (6/6) $p < 10^{-4}$	+3.02 **pp** (6/6) $p < 10^{-4}$	+3.82 **pp** (6/6) $p < 10^{-4}$	+2.69 **pp** (6/6) $p < 10^{-4}$	+3.86 **pp** (6/6) $p < 10^{-4}$

Table 6. Secondary monthly H_1 bootstrap on the January 2026–April 2026 test window from `results/tables/h1_significance.csv` (B = 1000, N = 4 monthly observations). Low statistical power by design; primary inference is in Table 5.

Hypothesis	$\widehat{\Delta \mathrm{SR}}$	95% CI	p
H_1^b: T2 vs T1	-0.075	[-8.575, +3.805]	0.578
H_1^c: T3 vs T1†	0.000	[0.000, 0.000]	1.000
H_1^{aux}: T1 vs B1	**+5.048**	[+2.691, +17.899]	**0.011**

† deployed T3 falls back to T1 (Section 4.3); the honest out-of-sample T3 *loses* to T1 (Table 7).

names, so the feature vector is unavailable and the policy *falls back to T1* on every block (`decision/t3_hazard.py`). Every "T3" row in a backtest was therefore T1. A previously-reported +7.03 bp "H1c" figure came not from the allocator but from an out-of-band hazard-ranking on a leaky full-panel fit; **we retract it**.

(ii) Wired and tested honestly, T3 loses. We (a) merged the real F1 lead features into the state so the Cox model genuinely evaluates, and (b) replaced the leaky full-panel fit with an *expanding-window* walk-forward — for each window, train strictly on prior blocks (purged by horizon+embargo), then evaluate out-of-sample. The honest result (Table 7) is a *loss*: T3 underperforms T1 by mean ΔAPY = -5.97 bp (95 % CI [-9.89, -2.76], 0/5 windows).

Table 7. **Honest out-of-sample T3 vs T1**, expanding-window walk-forward (train strictly pre-window; T3's Cox model actually executed). T3 loses on every window; mean ΔAPY = -5.97 bp. W1 is dropped (no pre-window data). Source `scripts/walkforward_t3_expanding.py`.

Window	Span	T1 APY	T3 APY	Δ bp
W2	Feb–Apr'25	6.92%	6.79%	−13.5
W3	May–Jul'25	7.80%	7.77%	−3.1
W4	Aug–Oct'25	7.17%	7.10%	−7.2
W5	Nov'25–Jan'26	5.87%	5.83%	−4.5
W6	Feb–Apr'26	5.61%	5.59%	−1.6
			mean	**−5.97**

The model-driven dwell estimate mis-times switches relative to T1's EWMA dwell. The scientific reading is clean: the F3 cross-protocol spread *is* the decision variable, the gas-aware threshold T1 captures it, and the hazard layer adds nothing out-of-sample — a pre-registered negative result that preempts the "if T3≡T1, why ML?" question.

Capacity (\$1M–\$50M). Krause-2005 yield-impact on the real-gas panel, under the *continuous* rate-depression model (the correct cost for at-par mint/burn lending — a per-rebalance round-trip slippage model double-counts and is superseded): T1 net APY eases from 8.57 % (\$1M) to 7.60 % (\$50M), retaining a +2.98–3.93 pp edge over passive Aave across the range. The binding constraint, however, is venue *depth*, not per-trade slippage: the edge-carrying venues Euler (TVL ∼ \$16M) and Spark (∼ \$29M) cannot absorb a \$25M+ deposit, so the realistic full-strategy capacity ceiling is ∼ \$5–10M; above it the allocator is effectively deep-venue-only, with a smaller but still positive edge. (The sweep runs on the deployed allocator, which is T1.)

4.4 Gas-sensitivity and real gas

The backtest uses a flat 25 gwei placeholder (real gas history was unavailable at submission time). We bound the resulting uncertainty two ways. First, a *sensitivity sweep* (`scripts/greedy_gas_sensitivity.py`) re-runs the policies at 10–200 gwei. T1's gas gate cuts its switch count as gas rises ($125 \to 52 \to 26 \to 9$ at 10/25/50/100 gwei), so its net APY degrades only gracefully ($5.10 \to 4.94 \to 4.53 \to 4.46$ %) and stays far above passive Aave (3.26 %). A naive greedy spot-APY baseline (switch to the best rate every block, *no* gas gate) instead holds 424 rebalances at *every* level and collapses from 4.72 % (10 gwei) to 2.03 % (50 gwei) to *negative* −1.26 % (100 gwei) and −7.63 % (200 gwei, vs T1's 4.13 % on 6 switches): the T1-minus-greedy gap widens monotonically from +39 bp to +1,176 bp. Event-time switching alone is therefore *not* the contribution — the gas-aware *throttle* is, and its value (only ∼ 1.5 bp in the near-zero-gas test window) grows to hundreds of bp in the high-gas regimes where naive rate-chasing turns ruinous. T2, which does not gas-throttle, rebalances ∼ 159 times at *every* level and collapses to 0.31 % at 200 gwei: only T1's explicit gate is meaningfully gas-aware. Second, we fetched *real* historical gas over the panel via `eth_feeHistory` on a free archive RPC (no key): mean 3.7 gwei, and only ∼ 0.3 gwei across the Jan–Apr 2026 test window — 50–100× below the placeholder, reflecting the post-Dencun collapse in L1 base fees. Re-running T1 with real forward-filled gas raises its edge over passive Aave to +2.25 pp (from +1.77 pp under the placeholder), with 311 rebalances costing only \$61. So the placeholder is a *pessimistic* bound; the honest net-of-gas edge is larger. In the current ultra-low-gas regime the gas gate is near-moot and the binding execution cost at size shifts to slippage/MEV (Section 6.); gas-awareness binds in high-gas spikes (2024-Q4 mean 18.5 gwei) where T1's throttling beats T2's naivety.

Table 8. Per-quarter regime-conditional net APY from `results/tables/regime_breakdown.csv`. T3 matches T1 for the same reason as in Table 4 — the F1+F3 Cox model fires on the same blocks T1 does over Jan–Apr 2026.

Policy	2026-Q1 APY	2026-Q2 APY
B1 Always-Aave	2.75%	4.81%
T1 Threshold	**4.39%**	8.37%
T2 OU stopping	4.35%	**8.37%**
T3 Cox (F1+F3)‡	4.39%	8.37%

‡ deployed T3 falls back to T1 (Section 4.3); honest out-of-sample T3 loses to T1 (Table 7).

4.5 Regime-conditional breakdown

Splitting the test window by quarter (Table 8) reveals the per-policy heterogeneity that the 4-month aggregate compresses. T1 dominates the stable Q1 regime ($\sigma_{\text{spread}} = 0.57$ pp); T2 dominates the mean-reverting Q2 regime ($\sigma_{\text{spread}} = 3.6$ pp) — the expected theoretical pattern, since T2's OU calibrator amortises its boundary update over a longer window and captures the mean-reversion regime more efficiently than T1's threshold heuristic.

The asymmetry is precisely what a Krause-style elasticity-of-substitution argument [8] predicts: the slow-rate-adjustment branch (Compound's kinked-rate response to utilization) lags the fast-rate-adjustment branch (Aave's variable borrow rate) by exactly the timescale on which T2's OU calibrator fits its mean-reversion rate κ. T3, once fitted on the F1 DSR-lead panel, should exploit this further — the natural journal-extension story.

4.6 Signal-class contribution

The MacKenzie Table 3.2 signal taxonomy mapping to F1 (lead-rate), F3 (fragmentation), and F4 (related instruments) is fully implemented in the `decision/features/` package and unit-tested against the research-side builders; F2 (order-book dynamics) is deferred per design-spec risk R1. The headline ablation (above) returns out-of-fold C-index 0.563 (F3 only), 0.582 (F1+F3), 0.563 (F3+F4), 0.582 (F1+F3+F4): F3 dominates as predicted by the cross-protocol cointegration literature [15], F1 DSR delta contributes the only non-F3 concordance lift (+1.9 pp), and F4 contributes ∼ 0 pp above F3 alone — the substantive negative finding for one of the three live MacKenzie signal classes in this DeFi-lending mapping. The binding inferential result is Table 5: T1 outperforms passive holds of all six in-scope protocols by +2.7–4.1 pp in all 6 windows ($p < 10^{-4}$ each, Holm-corrected); the smallest-margin Euler V2 contrast (+2.69 pp) is the least robust and panel-dependent.

5. Discussion: Cross-Domain Transfer and the DeFi-as-HFT Hinge

Section 4. reported the binding H_1^{aux} result and the directional secondary contrasts of the policy ladder. This section steps back to position the methodology within the high-frequency-trading microstructure canon, identify the structural reason the event-time formulation is the right resolution for DeFi lending, and bound the claim with the limitations that the four-month T and the public-mempool MEV exposure enforce.

The central conceptual move of this paper is the recognition that the on-chain analog of MacKenzie's regression-based "squashing" of microstructure signals [6, p. 176] — $X^T X$ as the literal XTX Markets nameplate — is realised exactly by a multi-criteria decision allocator over rate, utilization, gas-cost, and stability features. Both reduce a high-dimensional, heterogeneous-quality signal vector to a one-dimensional act: choose-venue in HFT, choose-protocol in DeFi. The author's prior work on limit-order-book mid-price prediction [23] introduced a domain-aware dual-branch recurrent architecture that decomposes the LOB signal into the $(\text{price}, \text{volume})$ pair plus microstructure context; the present paper transposes the same compositional spirit to DeFi lending — the domain-natural decomposition pair is $(\text{spread}, \text{gas regime})$ plus the F1/F4 conditioning columns — but replaces the deep recurrent forecaster with a Cox proportional-hazards regression on event-time triple-barrier labels, which is the model class the per-block label structure actually admits. The multicriteria precedent in crypto-portfolio selection [21] supplies the weighting-scheme idiom; this paper supplies the gas-aware threshold gate that makes the aggregation actionable at per-block resolution.

5.1 Signal-class taxonomy and the F1–F4 mapping

[6, Table 3.2, p. 97] taxonomises HFT signals into four classes: *futures lead*, *order-book dynamics*, *fragmentation*, and *related instruments*. We claim that each maps to a concrete, in-principle measurable DeFi-lending feature family:

F1 (lead). The Maker DSR redemption rate, sDAI exchange rate, and Curve 3pool USDC/USDT/DAI swap rate move ahead of Aave and Compound USDC supply rates with a lag of roughly six hours, by the elasticity-of-substitution mechanism that [8] documents for closed-form market-depth substitution in unified-pool systems. Source: subgraph events, free. Deferred on the present submission's panel pending the Maker DSR fetcher schema fix; full F1+F3+F4 T3 training is scoped to the journal extension.

F2 (order-book dynamics). Pending mempool transactions targeting each pool — deposits, withdrawals, borrows, repays — are observable *before* they execute, exactly as a Glosten–Milgrom dealer observes a quote-revision intent before the trade [7, Ch. 3]. This is the most direct analog of the ATD "Goldman leaves its bid" signal that [6] describes on pp. 102–104. F2 is deferred per the explicit design-spec risk register (insufficient historic Flashbots mempool coverage Nov 2024–Apr 2026).

F3 (fragmentation). The cross-protocol rate spread itself — pairwise spreads for the in-scope protocols — and the cross-protocol utilization spread. This **is** the decision variable, and the F3-only T3 fit on the submission panel returns an out-of-fold Cox C-index of 0.563 (rising to 0.582 once F1 is added) with the `f3_dispersion_std` coefficient dominant ($\hat{\beta} \approx +60$) and the `f3_spread_top2` coefficient large and negative ($\hat{\beta} \approx -20$), matching the physical intuition: high cross-protocol APR dispersion shortens regime dwell, large top-vs-runner-up gap lengthens it. [15] reports a Compound-leads-Aave cointegration with a 0.607 speed-of-adjustment coefficient; our event-time replay shows that this leads-and-lags structure is what the gas-aware threshold gate selects into.

F4 (related instruments). ETH spot price (drives gas regime), top-LP concentration per pool (more transparent on-chain than in MacKenzie's anonymous order books), and USDC/USDT/DAI peg deviations as leading indicators of liquidity stress; the [16] cross-chain comparative analysis catalogues the relevant risk-parameter levers. Deferred on this submission's panel.

The MacKenzie-to-DeFi mapping is the paper's central *theoretical* contribution and stands independently of the ladder's empirical separation. The Kissell impact-vs-edge inequality [9] supplies the gating threshold that converts each of F1, F3, and F4 into an actionable trigger.

5.2 The hinge: allocator + protocol launches as mutually reinforcing

Andrew Abbott's notion of a *hinge*, recapitulated by [6, pp. 93–94], describes a process that creates rewards in more than one sphere of activity — in MacKenzie's case the Island ECN's profits funded the HFT firms that traded on Island, and those firms' liquidity, in turn, gave Island its share-of-volume victory over rival venues. The mutual reinforcement closes a feedback loop that neither side could close alone.

The DeFi analog is sharper than the HFT original. A multi- protocol gas-aware allocator's fragmentation-rent simultaneously **(a)** earns the allocator alpha, and **(b)** makes it rational for a new lending protocol to launch — because differentiated-rate venues now attract sophisticated flow that bootstraps TVL, which feeds back into the rate-quote signal that the allocator profits from [15]. The empirical evidence is the protocol ordering visible in Table 4: Morpho Blue and Euler V2 both launched after the 2024-Q4 baseline window, and each is meaningfully on the Pareto frontier of (yield, TVL, switching cost) for at least one of the test-window quarters reported in Table 8. The pre-condition that [6, p. 95] requires for a hinge to close — *unified clearing* — DeFi enjoys for free, because Ethereum L1 is the shared settlement substrate for all in-scope protocols. Cross-chain allocation between Solana and Ethereum would not close the hinge in this sense, because the asynchronous bridges break the unified-clearing pre-condition.

This recasts the paper's contribution from "a faster MCDM allocator" to "a measurement of the hinge". The +5.048 Sharpe-unit gap that T1 captures over passive Aave hold (Table 5) is the empirical magnitude of the hinge-rent that an event-time allocator can extract from the multi-venue equilibrium. The gas-cost-crossover inequality of Section 3.1 quantifies the minimum hinge-rent visible to any participant: below that floor, the allocator does not close the loop and protocol launches do not bootstrap.

5.3 Limitations: MEV exposure and the asymmetric-speed-bump fix

The backtest does not deduct miner-extractable-value (MEV) losses on the rebalance transaction — a family named by [24] and measured at L1 scale [25, 12]. Public-mempool submission of a \$1 M rebalance would face an expected 5–30 bp sandwich tax, which would erode much of the T1 edge documented in Table 4. The reference deployment implementation therefore routes rebalances through the Flashbots private mempool via `eth_sendPrivateTransaction`, as specified in Section 3..

We reframe this protection in MacKenzie's terms. [6, pp. 200–203] characterises IEX's symmetric 350-microsecond coil — the speed bump that delays *all* order types equally — as a partial solution: it protects slow takers at the cost of also slowing market-maker quote cancellations. The Flashbots private mempool is structurally different: it delays *visibility* to MEV bots until inclusion — an asymmetric speed bump in MacKenzie's sense — while the allocator's own intent-cancellation and re-pricing remain fast. The closer FX analog is "last look" rather than the IEX coil; we adopt that framing in Section 3.1.

The second limitation is the $T = 4$ monthly Sharpe sample size. With $T = 12$ (one calendar year), the $\sqrt{T-1}$ pre-factor in the Lopez de Prado DSR formula [10, Ch. 14.7.3] would relax by a factor of $\sqrt{11/3} \approx 1.92$ and the secondary-contrast gate would become substantially easier to clear; the conservative posture is deliberate and traceable to the panel's coverage start at 2024-11-01. The third limitation is the deferred F2 mempool channel: once Flashbots historic snapshots stabilise, the $X^T X$-style regression of [6, p. 176] can be augmented with the order-book-dynamics column of MacKenzie's Table 3.2.

The methodology contribution — event-time decision frame, F1–F4 signal taxonomy mapping, hinge measurement, and reproducible fetcher and replay infrastructure — stands independently of the ladder's secondary-contrast outcomes. The allocator-as-XTX bridge to [6] and the hinge-measurement reframing of the contribution are the durable deliverables; the H_1^{aux} outperformance of passive Aave hold is the empirical anchor on which they rest.

6. Limitations and Threats to Validity

We disclose every methodological constraint affecting the headline results so readers can scope conclusions appropriately and so that reproductions can target the same envelope.

(i) Walk-forward N is small (5–6 windows). The binding T1-vs-holds contrasts are significant ($p < 10^{-4}$ on all six, Holm-corrected), but per-window ΔAPY standard errors are $\sim$ 5–10 bp and CIs are wide at this N; overlapping windows with an overlap-corrected bootstrap would tighten them (journal extension). The honest out-of-sample T3-vs-T1 result (-5.97 bp, Table 7) is a clear negative at this N.

(ii) Gas placeholder (now bounded). The headline backtest uses a constant 25 gwei; Section 4.4 bounds this with a 10–200 gwei sensitivity sweep and with real `eth_feeHistory` gas (test window $\sim$ 0.3 gwei, 50–100× lower), confirming the placeholder is pessimistic and the gas-aware gate is the binding mechanism only in high-gas regimes. The F4 `gas_log10` sub-signal is still dropped as zero-variance under the constant; the zero-variance filter drops the column, leaving F4 with only ETH/USD and USDC peg. F4 thus contributes *zero* OOF C-index above F3-only — a methodological, not empirical, finding. Real per-block gas re-run is queued.

(iii) F2 mempool deferred. The MacKenzie F2 "order-book dynamics" class requires paid Blocknative or archived mempool ($\sim$ 30 TB); not available in this study.

(iv) Heterogeneous rate-refresh cadence. T1/T2/T3 actively switch among all six protocols on the per-block grid, but the six refresh their on-chain rates at different native cadences: Aave V3 ($\sim$ 48 s) and Euler V2 ($\sim$ 6 min) are genuinely sub-hourly; Spark and Compound V3 are sub-hourly via subgraph and RPC sampling; Morpho Blue is ∼hourly; and Fluid updates only ∼daily. Switch decisions into the three slower venues act on a forward-filled most-recent rate (the live-agent information set), so we claim a per-block *decision loop*, not per-block rate *resolution* for all six. The three genuine sub-hourly venues carry $\sim$ 63–64% of held time (Table 3); per-block fetchers for the slower three are the next-extension scope.

(v) Capacity is analytic. The capacity sweep applies the linearized Krause-2005 yield-impact rather than re-running the replay engine at each position size; the time-share-weighted average smooths the small-pool tail.

(vi) Prospective deployment validation. The evaluation reported here is retrospective and testnet/paper-trading only. A prospective study is required before drawing conclusions about execution quality or realized performance in production markets.

Code and data availability. All artefacts are public. End-to-end reproducible pipeline (data extension, T3 training, walk- forward, capacity, paper rebuild) plus walk-forward equity parquets, T3 Cox-hazard training artefacts (per-ablation sidecars, `results/models/t3_cox_*.json`), and supplementary reproducibility materials: `https://github.com/SergeySolovyev/icicpe-2026-event-time-mcdm-defi` (`python -m scripts.finalize_6way_pipeline`). The per-block 6-way panel (∼72 MB parquet, Nov 2024–Apr 2026) is mirrored on HuggingFace Datasets at `https://huggingface.co/datasets/sssolovjov/icicpe-2026-defi-panel`. Reference implementation of the deployment loop (per-block execution, private relay support, observability endpoints, and an append-only audit trail): `https://github.com/SergeySolovyev/event-time-mcdm-agent`.

7. Conclusion

We specified an event-time, gas-aware multi-protocol DeFi lending allocator with three nested decision-policy tiers (T1 threshold, T2 OU optimal stopping, T3 Cox proportional-hazards), each grounded in a distinct microstructure or quantitative-finance source. On a 3,931,200-block panel covering January 2026–April 2026 across 6 active Ethereum USDC lending venues, T1 beats passive Aave V3 hold by +5.048 Sharpe ($p = 0.011$). We then report a *pre-registered negative result*: the F1+F3 Cox T3 tier, trained strictly out-of-sample and actually executed, does *not* beat T1 — it loses by -5.97 bp on the honest expanding-window walk-forward. Real `eth_feeHistory` gas confirms the net-of-gas T1 edge (+2.25 pp over passive Aave) exceeds the conservative placeholder backtest.

The contribution stands at three layers. **Methodology**: event-time gas-aware decision rules admit two to three orders of magnitude more rebalance opportunities than hourly polling, and the four-class HFT signal taxonomy of [6] operationalises cleanly on multi-protocol DeFi lending. **Infrastructure**: six-way event-stream fetchers, a per-block replay engine, and public data and code artefacts support independent replication. **Empirical**: the binding finding complements the equilibrium impossibility result of [14] by demonstrating that the residual exploitable edge in DeFi lending arises in the inter-block microstructure — precisely where the HFT taxonomy predicts it should.

Future work should extend the F1 lead-rate features to Maker DSR and Curve 3pool at per-block cadence, reestimate T3 with observed gas costs, and incorporate F2 mempool features once sufficiently complete historical coverage is available. A separate prospective validation study should then evaluate execution quality in a controlled testnet or paper-trading setting before any claims are made about mainnet deployment.

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